

DTMC

Chance

1. PCTL formulas are ϕ with the following syntax:

$$\begin{aligned}\phi &::= \mathbf{true} \mid p \mid \phi_1 \wedge \phi_2 \mid \neg\phi \mid \mathbf{P}_{\sim p}[\psi] \\ \psi &::= \mathbf{X}\phi \mid \phi_1 \mathbf{U}^{\leq k} \phi_2 \mid \phi_1 \mathbf{U} \phi_2\end{aligned}$$

where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

- (a) Give the formal semantic of PCTL, interpreted over DTMC.

Answer: Below (see also the slides on DTMC).

Let $M = (S, s_0, P, L)$ be your DTMC. A PCTL formula ϕ holds on M if ϕ holds on the initial state s_0 , denoted by $s_0 \models \phi$. We now give meaning of $s_0 \models \phi$, for any state s . See below:

$$\begin{aligned}s \models p &= p \in L(s) \\ s \models \neg\phi &= \text{not}(s \models \phi) \\ s \models \phi_1 \wedge \phi_2 &= s \models \phi_1 \text{ and } s \models \phi_2\end{aligned}$$

For the probabilistic operator $\mathbf{P}_{\sim p}[\psi]$ the definition is as below. I only give the def. for ' $\leq$ ' as the $\sim$. The definition is similar for other cases of $\sim$.

$$s \models \mathbf{P}_{>q}\psi = \underbrace{P\{\omega \in Paths(s) \mid \omega \models \psi\}}_{\text{probability}} > q$$

Here, $Paths(s)$ is the set of all possible paths/executions that starts from s ; typically there are infinitely many of them. The notation $\omega \models \psi$ means that the (temporal) property ψ holds on the path ω .

Finally, the notation $P\{\omega \in Paths(s) \mid \omega \models \psi\}$ refers to the probability that any such path ω occur if we are in the state s .

The meaning of $\omega \models \psi$ is defined recursively over the syntax of ψ as follows (the def. is not complete, you can see the slides for the rest):

$$\begin{aligned}\omega \models \mathbf{X}\phi &= \omega[1..] \models \phi \\ \omega \models \phi_1 \mathbf{U} \phi_2 &= \omega \models \phi_2 \text{ or } (\omega \models \phi_1 \text{ and } \omega[1..] \models \phi_1 \mathbf{U} \phi_2)\end{aligned}$$

- (b) Let's introduce some derived operators with the 'usual' meaning: $\vee$, $\rightarrow$, and $\mathbf{F}$ (or $\Diamond$). Propose a definition for them.

Answer: This can be defined as 'usual':

$$\begin{aligned}\phi_1 \vee \phi_2 &= \neg(\neg\phi_1 \wedge \neg\phi_2) \\ \phi_1 \rightarrow \phi_2 &= \neg\phi_1 \vee \phi_2 \\ \mathbf{F}\phi &= \mathbf{true} \mathbf{U} \phi\end{aligned}$$

- (c) Developer Peter Parker claims that $\neg\mathbf{P}_{>c}\phi$ is equivalent to saying $\mathbf{P}_{\leq c}\phi$. Is Peter right?

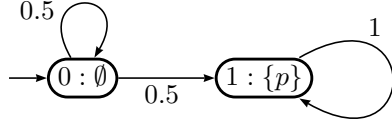
Answer: see in the slides :)

- (d) The 'usual' way to define $\mathbf{G}$ is: $\mathbf{G}\phi = \neg\mathbf{F}\neg\phi$. So, for example $\mathbf{P}_{\geq 0.8}[\mathbf{G}a]$ can be expressed as $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$. Unfortunately, the syntax of PCTL does not have negation over the temporal formula (we only have negation over proposition and probabilistic formulas). So, the formula " $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$ " is not allowed in PCTL. How do you then propose to define $\mathbf{P}_{\geq p}[\mathbf{G}\phi]$?

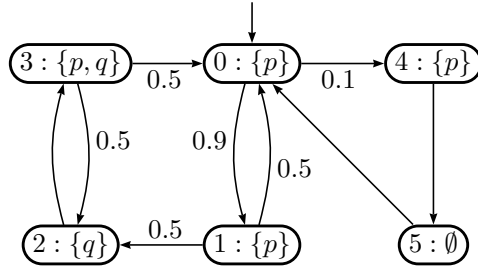
Answer: How about like this: $\mathbf{P}_{\leq p}[\mathbf{F}\neg\phi]$?

2. Explain why in PCTL (e.g. interpreted over DTMC) $\mathbf{P}_{\geq 1}[\mathbf{F}\phi]$ is not the same as CTL $\mathbf{AF}\phi$ (the former is weaker than the latter). Give an example that shows the difference. Note: $\mathbf{F}\phi$ is defined as *true* $\mathbf{U} \phi$.

Answer: Consider the DTMC M below. The CTL property $\mathbf{AF}p$ is not a valid property of M , because it can cycle in $S0$ forever, hence never reaching the state $S1$ where p holds. However, from PCTL perspective, the probability that M cycles in $S0$ forever would approach zero. And therefore the probability of eventually going over to $S1$ approaches 1.



3. Consider the following DTMC M modelling some system:



Consider the following PCTL properties to check:

- $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{< 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{=?} [p \mathbf{U} q]$

What do these formula say? Then describe the procedure to check them (or at least one of them) on M .

Answer:

The property $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$ is valid on a state s if the execution that starts from s would satisfy the path formula $\psi = p \mathbf{U} q$ with the probability of at least 0.82. When the reference state s is not mentioned, we mean it to be M 's initial state. Since all executions of M starts from its initial state, this is the same as saying that M 's executions satisfy the asserted (probabilistic) property.

The formula $\mathbf{P}_{=?} [p \mathbf{U} q]$ is a quantitative query (so, not a Y/N-query), asking what the probability of "a state s " to satisfy the path property $p \mathbf{U} q$. If no reference state is mentioned, then we mean to ask that on s_{init} (well, M 's initial state).

Also notice that answering the query $\mathbf{P}_{=?} [p \mathbf{U} q]$ for every state s in M will enable us to answer the other two queries/formulas above.

Let's now focus on the last formula; let's call the inner formula ψ :

$$\mathbf{P}_{=?} [\underbrace{p \mathbf{U} q}_{\psi}]$$

Let us now calculate, for every state, the probability that it satisfies ψ . We will proceed as follows:

- (a) We need to first perform "precomputation", to identify the so-called $\mathbf{S}^{yes}$ and $\mathbf{S}^{no}$ sets. We start with $\mathbf{S}^{no}$. This is the set of states of M on which $\mathbf{P}_{\leq 0}[\psi]$ holds. From the picture you can 'see' that this set is $\{4, 5\}$, but let us see how we can calculate this systematically.

Calculate first the set E , which is all the states satisfying the (non-probabilistic) CTL formula $\mathbf{E}[p \mathbf{U} q]$. You can do this e.g. with CTL standard model-checking. We end up with $E = \{0, 1, 2, 3\}$. Any state in E has a path satisfying ψ , and hence a non-zero probability of satisfying ψ . So, $\mathbf{S}^{no}$ must be $S/E = \{4, 5\}$.

The set $\mathbf{S}^{yes}$ consists of all states on which $\mathbf{P}_{\geq 1}[\psi]$ holds. To calculate this set, we first calculate the set F of all states t that have a path τ , starting from t , leading to $\mathbf{S}^{no}$, while maintaining the property $p \wedge \neg q$ along τ (up to entering until $\mathbf{S}^{no}$). We end up with $F = \{0, 1, 4, 5\}$ ¹. By its definition/construction, states in F has a non-zero probability to satisfy $\neg\psi$. So, $S/F = \{2, 3\}$ is the set $\mathbf{P}_{\geq 1}[\psi]$ we look for²

So, to summarize we have calculated:

- $\mathbf{S}^{no} = \{4, 5\}$.
- $\mathbf{S}^{yes} = \{2, 3\}$.

- (b) We can now calculate the probability of each state in M for satisfying ψ .

We can conclude that the probability of any state in $\mathbf{S}^{no}$ for satisfying ψ is simply 0. Conversely, the probability of any state in $\mathbf{S}^{yes}$ for satisfying ψ is 1.

For any state t that do not belong to either $\mathbf{S}^{yes}$ or $\mathbf{S}^{no}$, its probability for satisfying ψ is some $0 < p < 1$. We have two such states: $\{0, 1\}$. Let x_s denote the probability that a state s satisfies ψ (which is $p \mathbf{U} q$). We will have the following relations/equations; and x_i 's are the solutions of the equations:

$$\begin{aligned} x_0 &= P_{0 \rightarrow 1} * x_1 + P_{0 \rightarrow 4} * x_4 \\ x_1 &= P_{1 \rightarrow 0} * x_0 + P_{1 \rightarrow 2} * x_2 \end{aligned}$$

where $P_{u \rightarrow v}$ denotes the probability of taking the transition to v , if we are in the state u .

Now, we know that $x_4 = 0$ and $x_2 = 1$. So the above simplify to:

$$\begin{aligned} x_0 &= P_{0 \rightarrow 1} * x_1 \\ x_1 &= P_{1 \rightarrow 0} * x_0 + P_{1 \rightarrow 2} \end{aligned}$$

Filling the the values of the P (from M 's picture) we obtain these equations:

$$\begin{aligned} x_0 &= 0.9x_1 \\ x_1 &= 0.5x_0 + 0.5 \end{aligned}$$

You can apply algebra to solve the equations. E.g. the first says x_0 is $0.9x_1$. If we fill this in for the x_0 in the 2nd equation we get:

$$x_1 = 0.5 * (0.9x_1) + 0.5$$

This leads to the solution that $x_1 \approx 0.909$. And $x_0 \approx 0.818$.

¹Notice that $\{4, 5\}$ are states from $\mathbf{S}^{no}$ itself. In general, this F will indeed always include $\mathbf{S}^{no}$.

²The MC-slides in the website of Prism, F is characterized differently, namely as the set of states where the probability of ψ is < 1 . This should be the same as saying that there is a non-zero probability of $\neg\psi$, which is how I looked at it above.

(c) Now we can answer all formulas/queries that were posed in the original question:

- $\mathbf{P}_{=?} [p \mathbf{U} q]$. On s_{init} the probability is 0.818.
- $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$. Well, then this is **not** valid on s_{init} .
- $\mathbf{P}_{< 0.82} [p \mathbf{U} q]$. This is valid on s_{init} .

4. Consider again the DTMC M in No. 3. On which states of M the following properties hold?

- $\mathbf{P}_{\leq 0} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 1} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 0.18} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{< 0.18} [p \mathbf{U} \neg(p \vee q)]$.

5. Consider again the DTMC M in No. 3. Which states satisfy $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \mathbf{P}_{\geq 1} [q \mathbf{U} p]]$? (Well, obviously state 5, but which other states, if any, satisfy the property?)

Answer: Let's first introduce some names to abbreviate:

$$\mathbf{P}_{\leq 0.5} [p \mathbf{U} \overbrace{\mathbf{P}_{\geq 1} [q \mathbf{U} p]}^{\chi}]$$

ψ

We will first calculate which states satisfy $\chi = \mathbf{P}_{\geq 1}[\psi]$. We proceed with the checking procedure as in the examples in (3):

- We calculate the set E of states satisfying $\mathbf{E}[q \mathbf{U} p]$, and obtain $E = \{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ for ψ is $\{5\}$ (the complement of E).
- Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg p$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ (for ψ) is the complement of F . So, it is $\{0, 1, 2, 3, 4\}$.
- From the above we conclude that the probability of satisfying ψ is 1 on all states, except on state 5, for which the probability is 0.
- So, the states in $\{0, 1, 2, 3, 4\}$ satisfy $\mathbf{P}_{\geq 1}[\psi] = \chi$.

Next, we proceed to calculate the probabilities of different states to satisfy $p \mathbf{U} \chi$. From this we can see which states would satisfy $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \chi]$ (which abbreviates $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \mathbf{P}_{\geq 1} [q \mathbf{U} p]]$).

The calculation proceeds in the same way:

- We calculate the set E of states satisfying $\mathbf{E}[q \mathbf{U} \chi]$. Treat χ as an atomic proposition. Previously we already calculated which states satisfy χ , namely $\{0, 1, 2, 3, 4\}$; so, add χ as a 'label' to these states. We then obtain that E is also $\{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ (for $q \mathbf{U} \chi$) is $\{5\}$.
- Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg \chi$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ is the complement of F . So, it is $\{0, 1, 2, 3, 4\}$.
- From the above we conclude that the probability of satisfying $q \mathbf{U} \chi$ is 1 on all states, except on state 5, for which the probability is 0.
- So, the only state satisfying the asked property $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \chi]$ is state 5.